E-Commerce Sales & Customer Behavior Analysis using Python

Exploratory Data Analysis (EDA) Report

Dataset: UCI Online Retail (December 2010 - December 2011)

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

Submitted by: **Anunay Kumar Sah**

B.Tech Computer Science, ITER (Siksha 'O' Anusandhan), Bhubaneswar

GitHub: github.com/aksah20

Project 01 of the Bleep EDA Project Assignment

1. Introduction and Objective

An online retail company wants to understand its sales performance, how its customers buy, which products are in demand, and where its customers are located. In this project I use Python to clean a real transaction dataset, explore it from different angles, and turn the findings into practical recommendations for the business.

The analysis tries to answer these ten questions:

- Which products generate the highest sales?
- Which countries contribute the most revenue?
- What are the monthly and weekly sales trends?
- Which products have the highest quantity sold?
- What is the average order value?
- Who are the most valuable customers?
- Are there unusual transactions or outliers?
- What percentage of transactions are cancelled?
- Which products show increasing or declining demand?
- Is there a relationship between product price and quantity purchased?

2. Dataset and Data Understanding

The dataset is the **UCI Online Retail dataset** (https://archive.ics.uci.edu/dataset/352/online+retail). It contains all transactions of a UK-based online gift retailer between 1 December 2010 and 9 December 2011. Many of the retailer's customers are wholesalers. The raw file has **541,909 rows and 8 columns**. Each row is one product line inside an invoice, not a full order, so I grouped rows by invoice number whenever I needed order-level numbers.

Column	Meaning
InvoiceNo	Invoice number. If it starts with C, the invoice was cancelled
StockCode	Product code
Description	Product name
Quantity	Units of the product in that row
InvoiceDate	Date and time of the invoice
UnitPrice	Price of one unit in GBP
CustomerID	Customer identifier
Country	Country of the customer

3. Data Cleaning

Before any analysis I checked the data quality and made the following decisions.

Issue found	Size	Decision
Missing CustomerID	24.9% of rows	Kept for sales, product and country analysis (still valid sales). Removed only for customer-level analysis.
Missing Description	1,454 rows (0.27%)	Removed. Mostly zero-price stock adjustments.
Exact duplicate rows	5,268 rows	Removed.
Cancelled invoices (start with C)	9,251 rows	Kept in a separate table for the cancellation analysis. Excluded from sales.
Negative quantity without C, or price of 0 or less	Small number of rows	Removed. These are stock adjustments, damaged goods or free items.
Non-product stock codes (POST, M, BANK CHARGES etc.)	Several codes	Removed. Real product codes start with 5 digits.
Two huge orders (80,995 and 74,215 units)	2 invoices	Removed. Both were cancelled by the same customer a few minutes later, so they were not real sales.

After cleaning, the sales data has **522,502 rows**, grouped into **19,771 orders** placed by **4,333 identified customers** from **38 countries** for **3,993 different products**. Total revenue is about **GBP 10.0 million**. I also created new columns: Revenue (Quantity x UnitPrice), YearMonth, Weekday and Hour.

Outlier analysis (Question 7)

Using the IQR method, many rows are flagged as outliers: 26,813 rows for quantity, 35,744 rows for unit price and 1,800 orders for order value. The largest remaining order is GBP 52,941 and the largest remaining single row is 4,800 units. These look like genuine bulk (wholesale) purchases and not mistakes, so I **kept them**. In the charts I used log scales or cut at the 99th percentile so they do not hide the normal behaviour. The only real errors were the two cancelled giant orders described above, which I removed. If I had kept them, they would have wrongly become the top products by quantity.

4. Exploratory Data Analysis

4.1 Distribution of quantity, price and order value

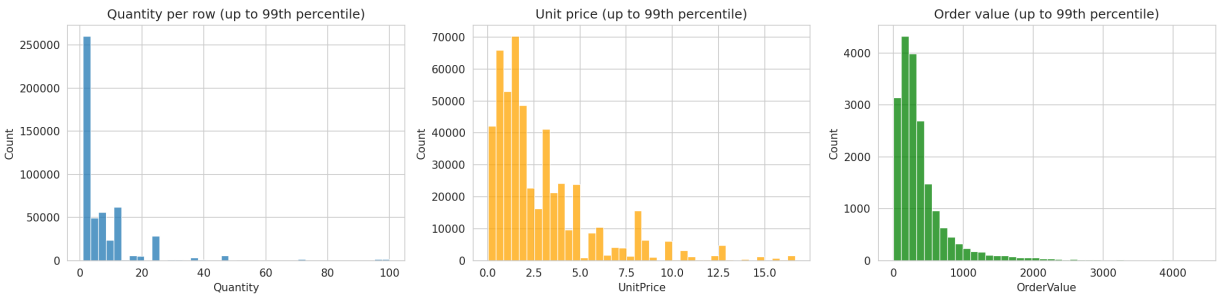


Figure 1: Distribution of quantity per row, unit price and order value (cut at the 99th percentile).

What this shows: All three variables are strongly right-skewed. The median row has only 4 units and a unit price of GBP 2.08, so most products are cheap and bought in small numbers. The median order is about GBP 302, but a few big orders stretch the tail to the right. The average order value (Question 5) is **GBP 505.85** while the median is **GBP 302.20**. The mean is much higher because of

wholesale orders, so the median describes a typical order better. An average order has about 273 units.

4.2 Top products by revenue and by quantity

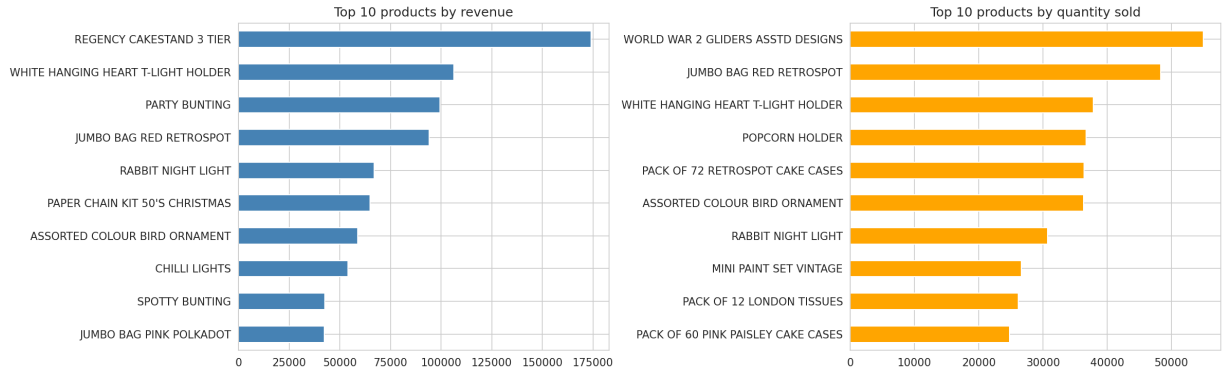


Figure 2: Top 10 products by revenue (left) and by units sold (right).

What this shows: The top revenue products are **Regency Cakestand 3 Tier (GBP 174,157)**, **White Hanging Heart T-Light Holder (GBP 106,237)** and **Party Bunting (GBP 99,445)**. By quantity, the leaders are **World War 2 Gliders (54,951 units)**, **Jumbo Bag Red Retrospot (48,371)** and the White Hanging Heart T-Light Holder (37,872). The two lists are different. The cakestand earns the most money without selling in the biggest numbers, while cheap items like the gliders win on volume. Jumbo Bag Red Retrospot and the T-Light Holder appear in both lists.

4.3 Revenue by country

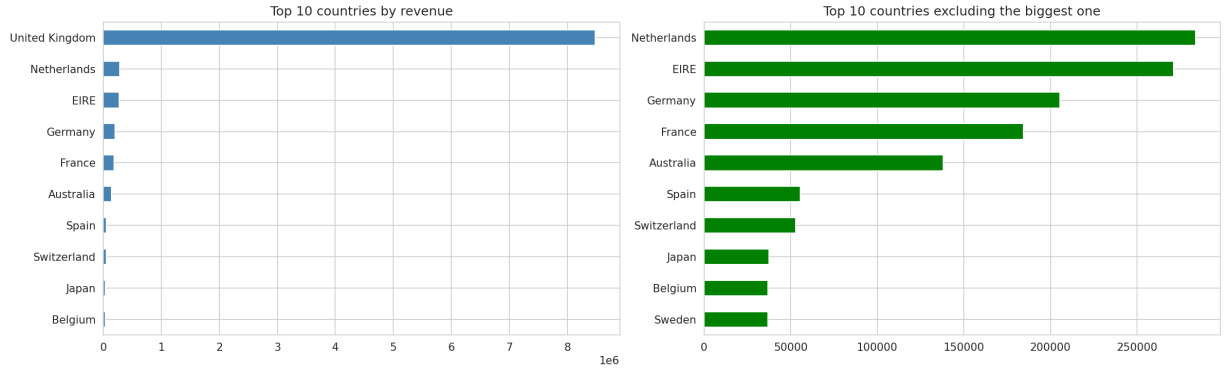


Figure 3: Top 10 countries by revenue (left) and the top export markets without the UK (right).

What this shows: The business depends heavily on its home market: the **United Kingdom gives 84.8%** of revenue and 90.5% of orders. The biggest export markets are the **Netherlands (2.8%), EIRE (2.7%), Germany (2.1%), France (1.8%) and Australia (1.4%)**.

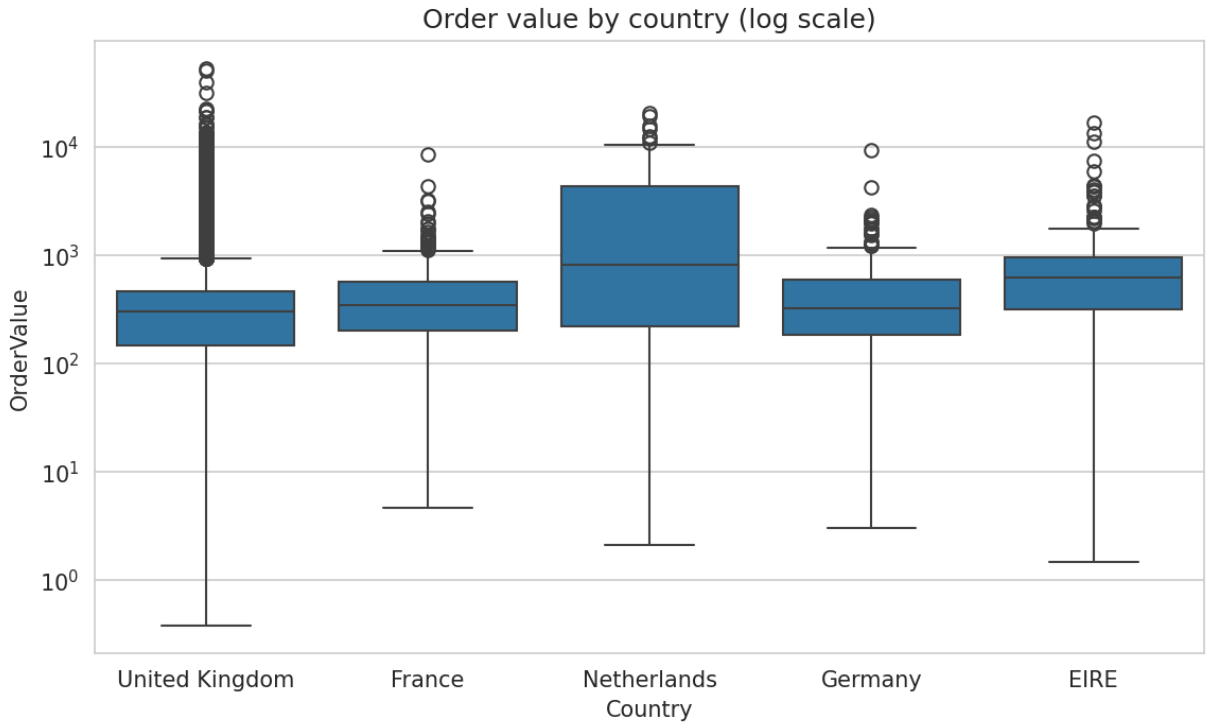


Figure 4: Order value by country for the five biggest markets (log scale).

What this shows: Orders from abroad are bigger. The median order is GBP 300 in the UK, but GBP 815 in the Netherlands and GBP 625 in EIRE. The average order value outside the UK is about GBP 813 compared with GBP 474 in the UK. This suggests that overseas customers are mostly wholesalers who place fewer but larger orders.

4.4 Sales over time

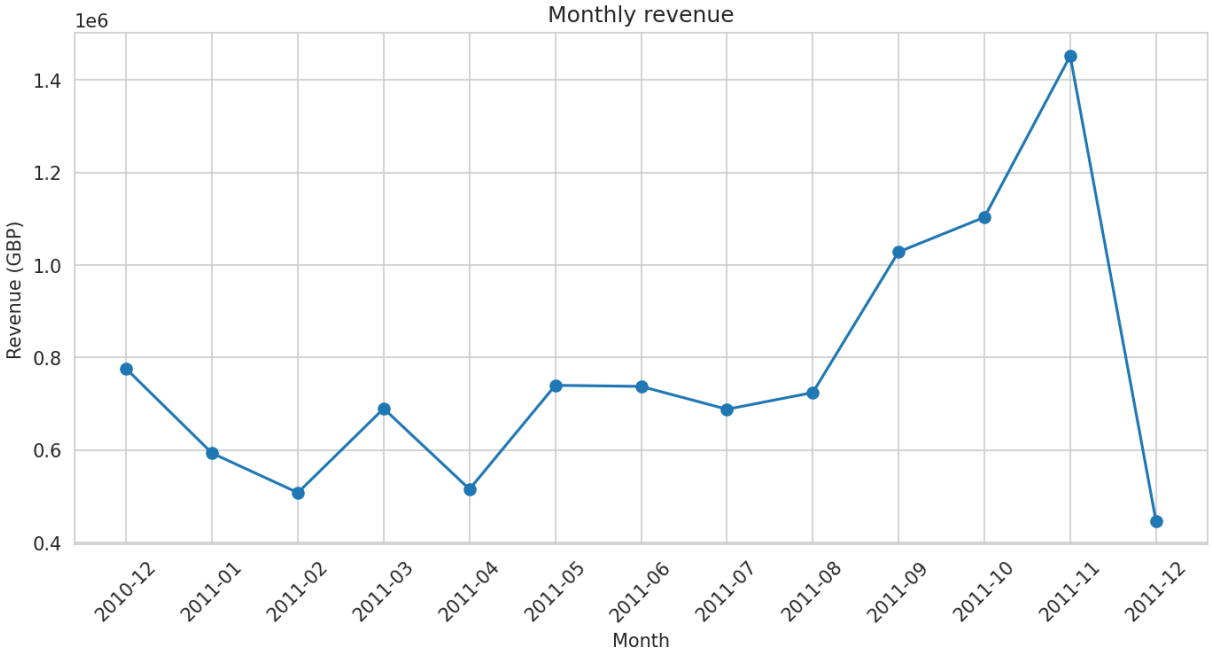


Figure 5: Monthly revenue.

What this shows: From January to August 2011 monthly revenue stays between about GBP 0.5 and 0.74 million. It then rises clearly: GBP 1.03 million in September, GBP 1.10 million in October and a

peak of **GBP 1.45 million in November**, about twice a normal month. This looks like the Christmas season, when retailers stock up. The drop in December 2011 is not real, because the data stops on 9 December, so that month is incomplete.

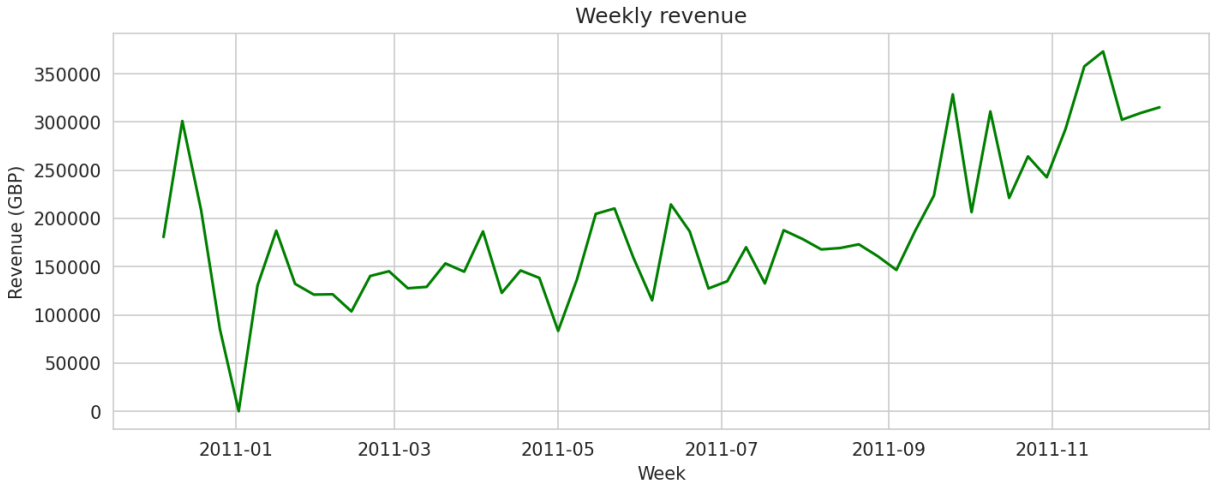


Figure 6: Weekly revenue.

What this shows: The weekly view shows the same pattern. From January to August weekly revenue stays between about GBP 83,000 and GBP 214,000, then it climbs from September and reaches its highest point of about **GBP 373,000 in the week ending 20 November**. There is also a clear gap around the new year: the data has no orders at all between 24 December 2010 and 3 January 2011, so one week shows zero revenue. This is probably the company's Christmas closure and not a real collapse in demand.

4.5 When do customers order?

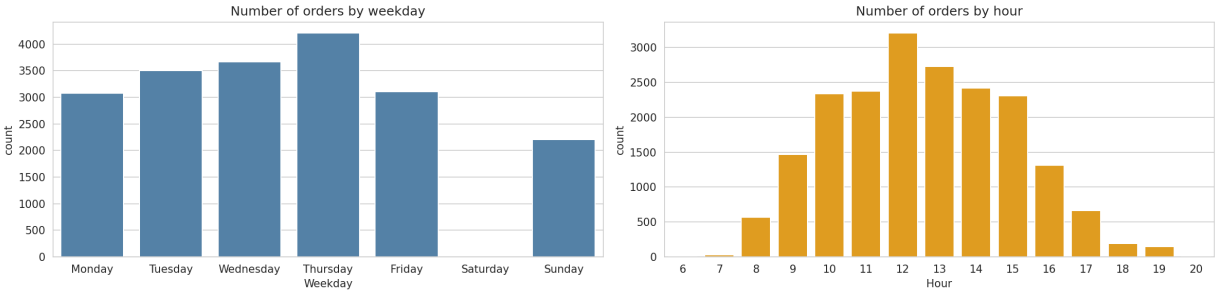


Figure 7: Number of orders by weekday (left) and by hour (right).

What this shows: Thursday is the busiest day with 4,209 orders, followed by Wednesday (3,667) and Tuesday (3,508). **There are no orders at all on Saturday**, and Sunday is the quietest day with 2,203 orders. During the day, orders start around 8 am, peak at **12 noon (3,204 orders)** and mostly happen between 10 am and 4 pm.



Figure 8: Number of orders by weekday and hour.

What this shows: The heatmap shows the same thing in one picture: orders are concentrated around midday on weekdays and are very low early in the morning and in the evening.

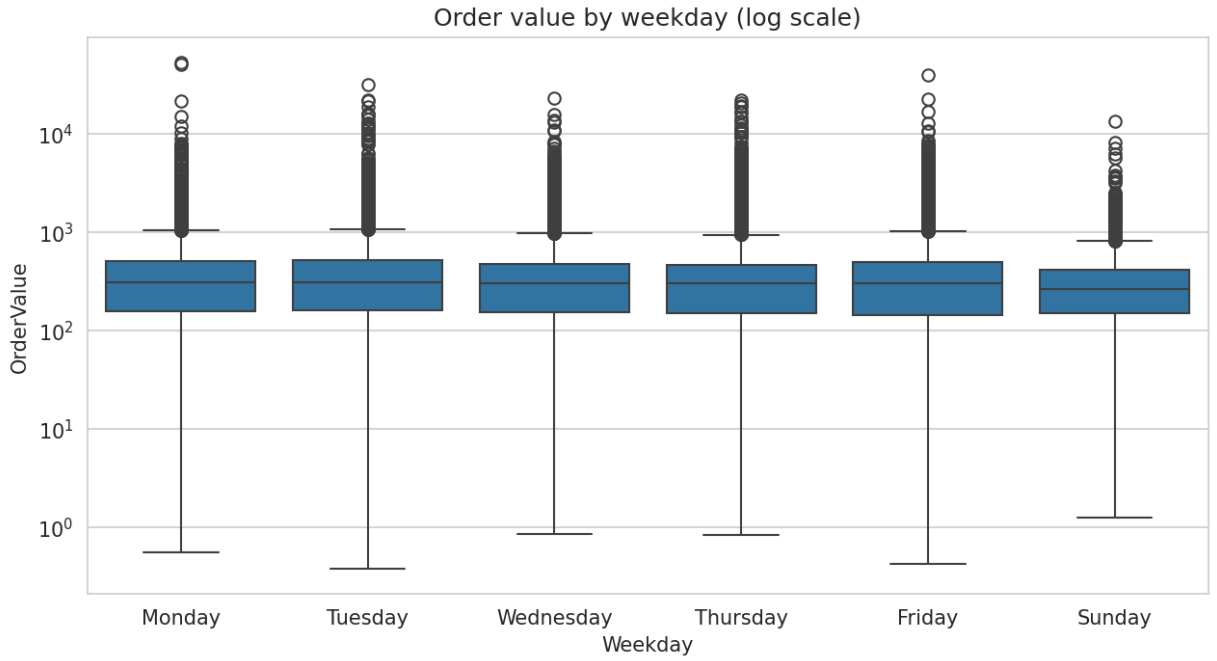


Figure 9: Order value by weekday (log scale).

What this shows: The median order value is almost the same from Monday to Friday (about GBP 301 to 309), so the weekday changes how many orders arrive, not how big they are. Sunday orders are a bit smaller (median GBP 262).

4.6 Customers

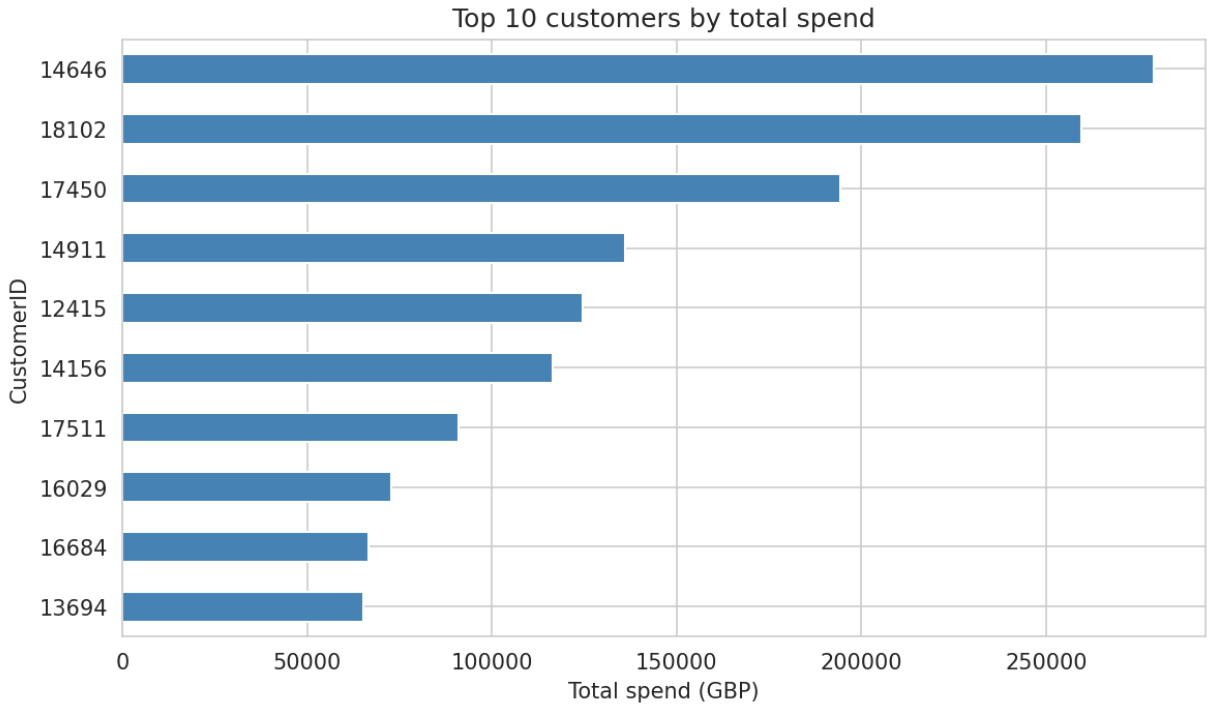


Figure 10: Top 10 customers by total spend.

What this shows: The most valuable customers are the ones with IDs 14646 (GBP 279,138), 18102 (GBP 259,657) and 17450 (GBP 194,391). The top 10 customers alone are responsible for about 16.6% of the revenue from identified customers.



Figure 11: Number of orders versus total spend per customer (log scales).

What this shows: Customers who order more often also spend more, and the relationship is strong (Spearman correlation 0.81). So repeat buying is the main driver of customer value.

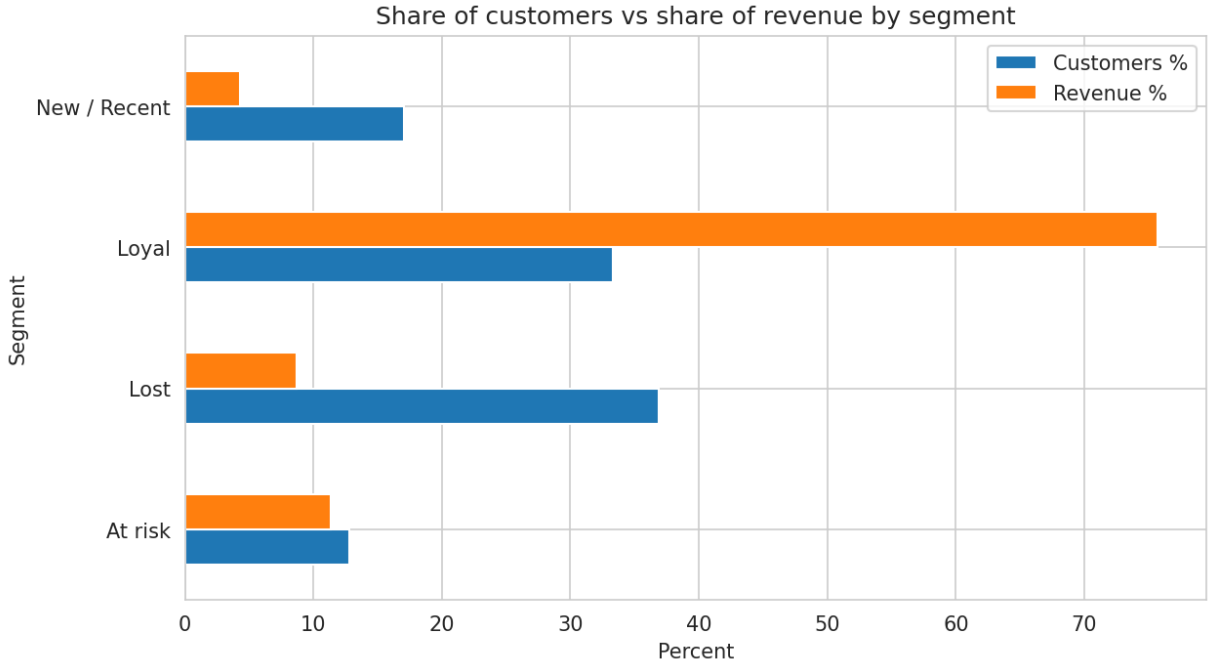


Figure 12: Share of customers versus share of revenue for each customer segment.

For segmentation I compared every customer with the median customer: the median customer last ordered 51 days before the end of the data and has placed 2 orders. A customer is 'recent' if they ordered more recently than the median and 'frequent' if they ordered more often than the median.

Segment	Customers	% of customers	% of revenue
Loyal (recent and frequent)	1,443	33.3%	75.7%
At risk (frequent but not recent)	555	12.8%	11.3%
New / Recent (recent but not frequent)	738	17.0%	4.3%
Lost (neither)	1,597	36.9%	8.7%

What this shows: A third of the customers (Loyal) bring in **75.7%** of the revenue, while the 'Lost' group is 36.9% of the customers but only 8.7% of the revenue. Overall, the top 20% of customers generate about **73.9%** of revenue, which is close to the 80/20 rule. The 'At risk' group has 555 customers who used to buy often and have spent about GBP 961,000 in total, so winning them back is worth the effort.

4.7 Correlation analysis

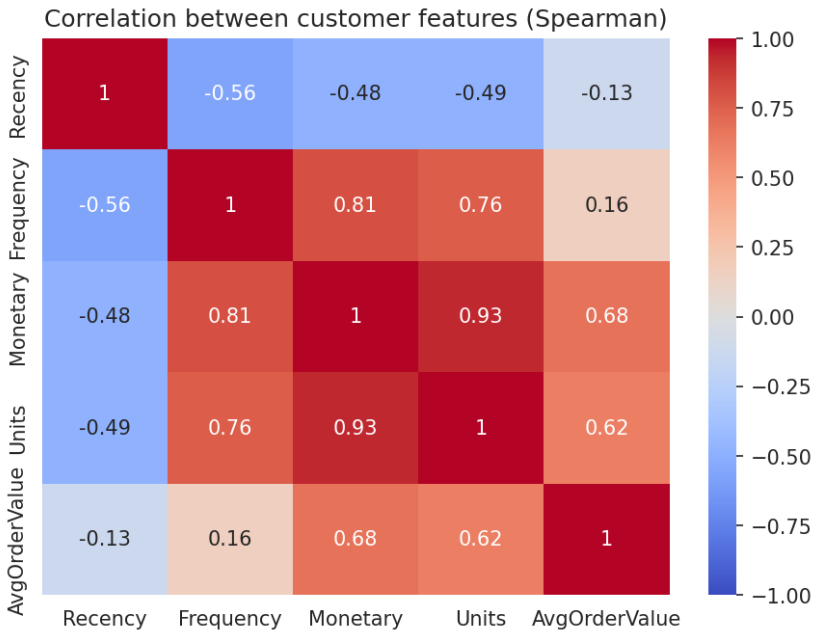


Figure 13: Spearman correlation between customer-level features.

What this shows: I used Spearman correlation because the data is very skewed and has outliers, and this method compares ranks instead of raw values. Frequency and total spend are strongly related (0.81), and total spend and total units are almost the same (0.93). Recency is negatively related to frequency (-0.56): customers who ordered recently also ordered more often. Average order value has only a weak link with frequency (0.16), so big spenders are mostly frequent buyers and not just people who place one big order.

4.8 Price versus quantity (Question 10)

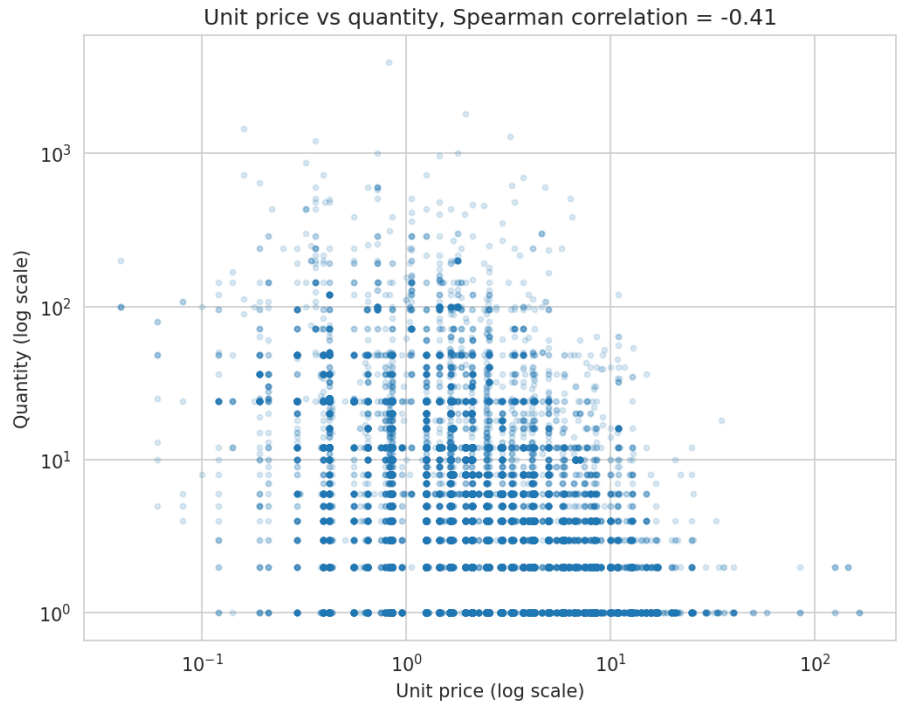


Figure 14: Unit price versus quantity per row (log scales, sample of 30,000 rows).

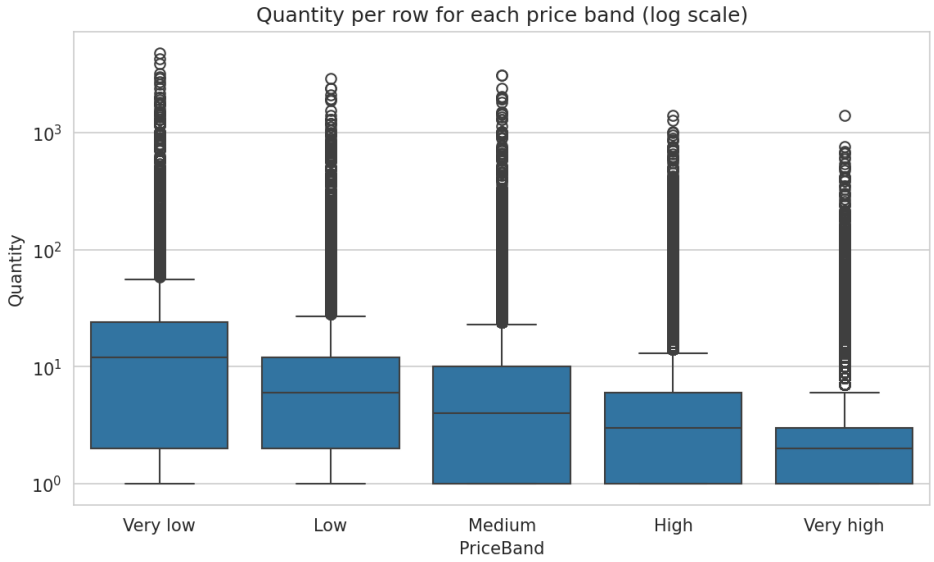


Figure 15: Quantity per row for five price bands (log scale).

What this shows: There is a negative relationship (Spearman correlation **-0.41**): the more expensive an item is, the fewer units are bought at a time. The median quantity falls step by step from **12 units** in the cheapest price band to **2 units** in the most expensive band (6, 4 and 3 in between).

4.9 Cancellations (Question 8)

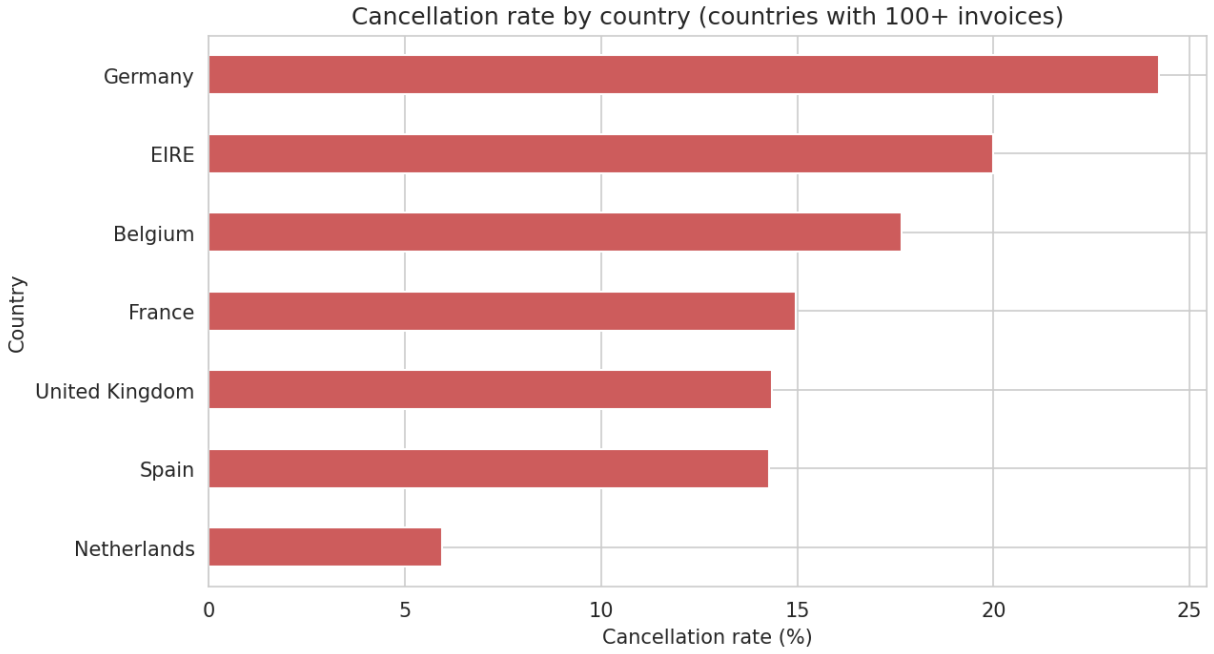


Figure 16: Cancellation rate by country (countries with 100 or more invoices).

What this shows: 14.8% of all invoices (3,836 of 25,900) are cancellations, and the cancelled goods are worth about **GBP 894,000**, which is about 8.9% of net sales revenue. Among countries with at least 100 invoices, the rate is highest in **Germany (24.2%), EIRE (20.0%) and Belgium (17.6%)**, compared with 14.4% in the UK. Some cancellations may belong to orders placed before December 2010, which are not in the data.

4.10 Product demand trends (Question 9)

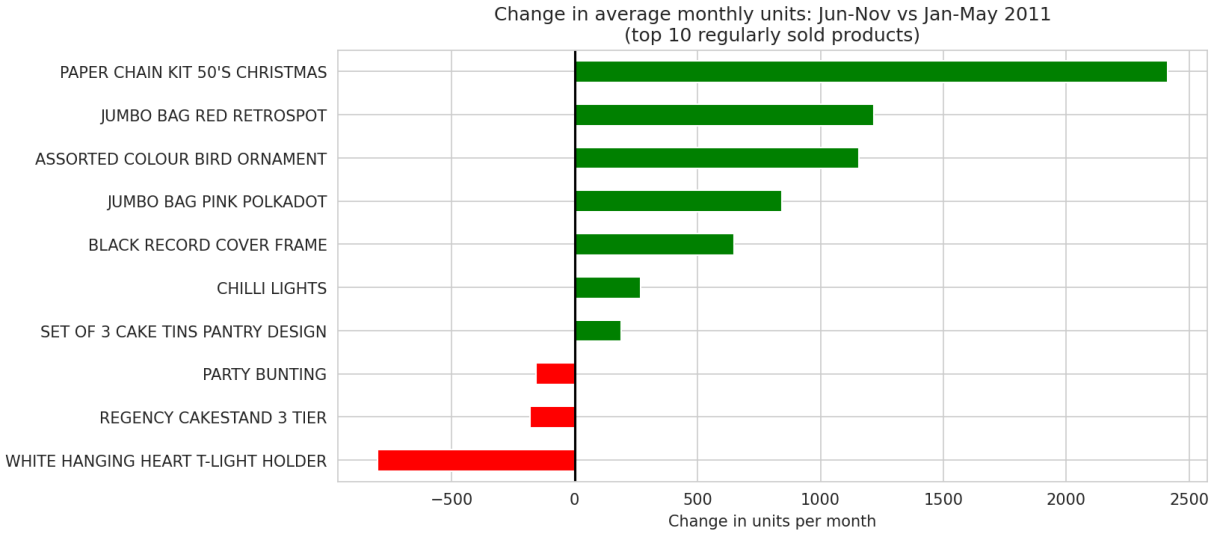


Figure 17: Change in average monthly units, June-November versus January-May 2011, for the top 10 regularly sold products.

What this shows: To check this fairly I used only full months (January to November 2011) and only products sold in at least 9 of those 11 months. The strongest growth is for the **Paper Chain Kit 50's Christmas (+2,411 units per month)**, which is clearly seasonal, followed by Jumbo Bag Red Retrospot (+1,218), Assorted Colour Bird Ornament (+1,156) and Jumbo Bag Pink Polkadot (+844). Demand fell for the **White Hanging Heart T-Light Holder (-802 units per month)**, the Regency Cakestand (-183) and Party Bunting (-157). The T-Light Holder and the Cakestand are among the top

sellers overall, so their decline is worth investigating.

5. Answers to the Ten Questions

#	Question	Answer
1	Highest sales products	Regency Cakestand 3 Tier (GBP 174,157), White Hanging Heart T-Light Holder (GBP 106,237), Party Bunting (GBP 99,445)
2	Revenue by country	UK 84.8%; then Netherlands 2.8%, EIRE 2.7%, Germany 2.1%, France 1.8%
3	Monthly / weekly trend	Flat from January to August (GBP 0.5-0.74M a month), strong rise from September, peak GBP 1.45M in November; December 2011 is incomplete
4	Highest quantity sold	World War 2 Gliders (54,951), Jumbo Bag Red Retrospot (48,371), White Hanging Heart T-Light Holder (37,872)
5	Average order value	GBP 505.85 (mean); GBP 302.20 (median)
6	Most valuable customers	IDs 14646, 18102 and 17450; top 20% of customers give 73.9% of revenue
7	Unusual transactions / outliers	Two giant orders (80,995 and 74,215 units) were cancelled and removed; the remaining outliers are genuine bulk orders (largest order GBP 52,941)
8	Cancellation percentage	14.8% of invoices, worth about GBP 894,000
9	Increasing / declining demand	Growing: Christmas paper chain kit, Jumbo Bag Red Retrospot, Bird Ornament. Declining: White Hanging Heart T-Light Holder, Regency Cakestand, Party Bunting
10	Price vs quantity	Negative relationship (Spearman -0.41); median 12 units in the cheapest band against 2 in the most expensive

6. Business Recommendations

- **Plan inventory for the autumn peak.** Revenue doubles from September to November, so stock up on seasonal and gift items from August and make sure bestsellers do not run out before November.
- **Protect and reward loyal customers.** One third of customers give about 76% of revenue. A loyalty or VIP programme (early access, bulk discounts) would help keep them.
- **Run a win-back campaign for 'At risk' customers.** 555 customers who used to buy often and spent about GBP 961,000 have gone quiet. A targeted email or discount is a cheap way to bring them back.
- **Target wholesale buyers abroad.** Customers in the Netherlands, EIRE, Germany and France place much bigger orders than UK customers. Bulk pricing and shipping offers can grow this segment, since the UK is already 85% of sales.
- **Time promotions for weekdays around midday.** Most orders arrive Tuesday to Thursday between 10 am and 4 pm, so emails and offers should go out just before. The company should also check why there are no Saturday orders (for example whether the site or warehouse is closed) and test if weekend sales are possible.
- **Use bundles for cheap products.** Cheap items are bought in large quantities (a median of 12 units) and expensive ones in small quantities (2 units). Multi-buy offers on cheap items and individual promotion of expensive items fit this behaviour.

- **Investigate the declining bestsellers.** The White Hanging Heart T-Light Holder and the Regency Cakestand are top earners but their monthly sales fell in the second half of 2011. Check for stock problems, price changes or new competition.
- **Reduce cancellations.** 14.8% of invoices are cancelled. Germany, EIRE and Belgium are well above the UK rate, so the company should look at shipping, product descriptions and order confirmation for those markets.
- **Improve data quality.** A quarter of the rows have no CustomerID, which limits customer analysis. Recording the customer for every order would give a much better picture of repeat buying.

7. Limitations

- The data covers only about one year, so I cannot confirm that the autumn peak repeats every year. December 2011 is incomplete.
- A quarter of the rows have no CustomerID, so the customer analysis (top customers, segments) covers only the identified customers.
- The retailer sells to many wholesalers, so the results may not apply to a normal shop selling to individual consumers.
- My customer segments use a simple median rule and are only one way to group customers.
- Some cancellations may belong to orders from before December 2010, so the cancellation rate is only approximate.
- This is exploratory analysis. Correlations show relationships, not causes, and I did not build a prediction model.

8. Conclusion

The analysis shows a business that is highly dependent on the UK market and on a small group of loyal, repeat customers, with a strong seasonal peak before Christmas and larger wholesale-style orders from abroad. Cheap items sell in big quantities, a few top products define the revenue, and about 15% of invoices are cancelled. Using these findings, the company can plan stock for the autumn peak, protect its loyal customers, win back customers at risk, grow wholesale sales abroad and reduce cancellations.

Resume description

Analyzed 500K+ e-commerce transactions using Python, Pandas, NumPy, Matplotlib and Seaborn to identify seasonal sales trends, top-performing products, geographic revenue distribution and customer segments, and recommended actions for inventory planning, customer retention and wholesale growth.